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Multivariate Extreme Spectral Bootstrap (SB) Procedure in Practice

TAE - Topics of Advanced Econometrics

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1 Introduction

Throughout the years, cryptocurrencies have continued to gain traction from not only experienced investors but also inexperienced investors that are only there for the hype. The massive trade volume of cryptocurrencies has introduced a new asset class characterized by extreme volatility and frequent black swan events. For risk management purposes, the modeling of extreme events, especially extreme losses, is critical. Traditional statistical methods, which rely on Gaussian assumptions, fail to capture the heavy-tail aspect of the financial returns, leading to underestimation of risk during crashes. This is even more true for financial assets such as cryptocurrencies which exhibits even heavier tail behavior compared to traditional financial assets. While Extreme Value Theory (EVT) provides a rigorous framework for modeling these rare events, the main issue lies in the lack of extreme observations, making inference inconsistent.

To address the challenge of data scarcity issue, bootstrapping techniques are often used to artificially increase sample sizes. However, standard bootstrap methods cannot be applied directly to EVT because the empirical distribution does not satisfy extreme value conditions. This creates a gap for when one tries to simulate stress scenarios for portfolio of assets that are both volatile and correlated. This is the motivation to apply the Multivariate Extreme Spectral Bootstrap (SB) procedure, recently proposed by Madhar, Legrand, and Thomas (2025). Relying on the spectral representation of Multivariate Generalized Pareto (MGP) distributions, it allows for the generation of synthetic extreme data that preserves the joint tail behavior of the original dataset.

This novel methodology is applied to a diversified cryptocurrency portfolio consisting of Bitcoin (BIT), Binance Coin (BNB), and Dogecoin (DOG). While the SB procedure has been demonstrated on traditional banking stocks, its application to the hyper-volatile crypto market presents a unique test of its robustness. The SB algorithm is applied to facilitate the estimation of key Tail Risk Metrics (TRM), specifically the Expected Shortfall (ES), Multivariate Marginal Expected Shortfall (MMES), and the Dependent Conditional Tail Expectation (DCTE).

The paper is structured as follows: After a short literature review on different methodologies and techniques developed on both bootstrapping and statistical methods for heavy tailed distributions, section 3 briefly introduces the methodology for generating synthetic data through the SB procedure with a discussion on the underlying assumptions, as well as a description of the data used for the portfolio construction. Section 4 shows the simulation results utilizing the SB procedure with the crypto portfolio. Followed by a discussion on the performance and potential improvements subject to digital assets in section 5.

2 Literature Review

To address the challenges of estimating the TRM of heavy-tailed distributions, which is critical for financial data, both traditional statistical methods relying on EVT as well as optimized bootstrapping procedures are explored. A persistent challenge across the literature is the scarcity of data in the tails, which complicates the estimation of the Extreme Value Index, high quantiles, and rare-event probabilities. The reviewed literature addresses these challenges through advanced resampling techniques with bootstrapping, bias-reduction strategies, and dynamic modeling for dependent time series. The main issues addressed in the papers can be categorized into four main areas: 1) bias-variance tradeoff and optimal order statistic selection (Gomes, Figueiredo, and Neves 2012; Gomes, Caeiro, et al. 2016; Qi 2008; Németh and Zempléni 2020), 2) serial dependence and heteroscedasticity in time series data due to volatility clustering thus violating the iid assumption (McNeil and Frey 2000; Bee and Trapin 2018; Davis et al. 2018), 3) data scarcity especially in the multivariate setting when joint extreme observations are rare (Madhar, Legrand, and Thomas 2025), and 4) the uncertainty in estimates due to the lack of tail information in the input model (Huang, Lam, and Liu 2023).

For bias correction, Gomes, Figueiredo, and Neves (2012) and Gomes, Caeiro, et al. (2016) focused on using second order parameters to reduce asymptotic bias without increasing the variance for the Classical-Variance Reduced-Bias estimators (CVRB). Németh and Zempléni (2020) proposed to use bootstrap samples and fitting a Generalized Extreme Value (GEV) distribution to the estimates then using linear regression to correct the bias of the Hill estimator. Qi (2008) studied how empirical likelihood and data tilting methods can be used to construct confidence intervals for tail indices and high quantiles.

On the topic of volatility clustering, methodologies focus mainly on data transformation and conditional modeling. McNeil and Frey (2000) proposed a two-stage approach by first fitting a GARCH model then applying EVT specifically the Generalized Pareto Distribution (GPD) on the tails of the residuals. Bee and Trapin (2018) reviewed several approaches including using quantile regression or allowing Peaks over Threshold parameters to vary over time based on autoregressive structures.

Although the approaches mentioned above address the difficulties in estimations and modeling of financial data in the extremes, the main issue of data scarcity in the tails that can be leveraged for model fitting and inference cannot be solved with the traditional statistical approaches alone. Bootstrapping is the main methodology used for addressing the lack of extreme data and the selection of optimal parameters. Both Qi (2008) and Gomes, Figueiredo, and

Neves (2012) focused on using double bootstrapping method for threshold selection to minimize the Asymptotic Mean Square Error (AMSE) of the Hill estimator. Davis et al. (2018) focused on block bootstrapping for time series with serial dependence to construct confidence intervals for the spectral tail process. Madhar, Legrand, and Thomas (2025) proposed a nonparametric spectral bootstrap procedure to create synthetic extreme data that preserves the asymptotic dependence structure of the underlying data in a multivariate case. Huang, Lam, and Liu (2023) discussed how heavy-tailed distributions are more vulnerable to truncation of the tails which would result in severe underestimation of rate-event probabilities. Standard bootstrap fails to quantify uncertainty in heavy-tailed inputs. However, bootstrapping assisted by GPD improves coverage.

The goal of this project is to extend the work done by Madhar, Legrand, and Thomas (2025) by applying the Multivariate Extreme Spectral Bootstrapping (SB) procedure to a more extreme case of the financial market, cryptocurrencies. Currently, Madhar, Legrand, and Thomas (2025) have demonstrated the consistency of their approach through simulated data and with market data with traditional bank stocks which are more stable. With the tail index ξ being between 0.4 to 0.6 for cryptocurrencies while traditional bank stocks with $\xi \approx 0.2$, the robustness of the SB procedure is tested.

3 Methodology

This section first discusses the theoretical framework behind the SB procedure and its underlying assumptions. Then a description of the underlying data used for the simulation and the rationale behind the choices is discussed.

3.1 Theoretical Framework

The SB procedure is a non-parametric joint simulation algorithm based on the spectral representation of Multivariate Generalized Pareto (MGP) vectors to generate samples from a MGP distribution. The algorithm requires n i.i.d. vectors, $Z_1, \dots, Z_n$, of standard MGP distribution in $\mathbb{R}^d$ where Z is defined as

$$Z = E + T - \max_{1 \leq k \leq d} T_k \quad (1)$$

E is the unit exponential variable and $T = (T_1, \dots, T_d)$ is the spectral random vectors defined by Rootzén et al. (2018).

For each $Z_{i,j}$ vector, $\Delta_{i,j}$ is calculated, which is defined as

$$\Delta_j = Z_j - \max_{1 \leq k \leq d} Z_k \quad (2)$$

Equation (1) can then be rewritten as

$$Z = E + \Delta \quad (3)$$

where $\Delta = (\Delta_1, \dots, \Delta_d)$.

$E_1, \dots, E_m \stackrel{iid}{\sim} \text{Exp}(1)$ are generated independently of Δ from historical data. And the bootstrapping is performed on Δ where m samples of $\Delta_1^*, \dots, \Delta_m^*$ are generated from $\Delta_1, \dots, \Delta_n$. The output of the algorithm is $m \in \mathbb{N}$ new i.i.d. copies of Z , denoted as $Z_1^*, \dots, Z_m^*$.

The SB procedure, in theory, should be even more suitable for cryptocurrencies than for traditional financial assets as it resolves the historical ceiling problem. This is critical for cryptocurrencies since record-breaking crashes is viewed more of a common feature rather than a market surprise. Cryptocurrencies also often crash together as they are closely tied to market regulations and exhibit high correlation during market downturns. However, the SB procedure requires the standardization of the historical data to exponential margins first, as can be seen from Equation (3). This is done under the assumption of a student-t distribution for the data. This assumption could be violated under the crypto context due to the fact that cryptocurrencies often exhibit skewed tails as crashes are often more severe than rallies. This could lead to a biased transformed data which would then lead to high error in the simulation. The underlying assumption of tail dependence, though theoretically true for cryptocurrencies, may still be violated as volatility of cryptocurrencies are also highly dependent on the hype of the coin itself. This is especially true if meme-coins are included in the investment portfolio.

3.2 Data Used

The portfolio for simulation consists of three cryptocurrencies: Bitcoin (BIT), Binance Coin (BNB), and Dogecoin (DOG). Five years of historical weekly prices from December 2020 to December 2025 are used for the simulation.

These three assets are purposefully chosen to ensure the critical assumption of tail dependence for the SB procedure. While they are often not correlated during bull runs, i.e. DOG spikes in price because of the hype from a tweet while BNB remains flat, they are highly correlated during bearish market conditions. The portfolio is also constructed in a way to mimic real behaviors of a crypto investor through the inclusion of both well known cryptos with high

trading volume, such as BIT and BNB, as well as meme coins that are randomly thrown into a portfolio due to unexpected hype on the internet, such as DOG. More formally, these assets are selected to represent distinct risk factors within the digital assets ecosystem: the market leader (BIT), exchange-utility tokens (BNB), and speculative meme assets (DOG).

4 Simulation Results

The performance of the SB procedure is evaluated by comparing its estimated TRM including Expected Shortfall (ES), Multivariate Marginal Expected Shortfall (MMES), and Dependent Conditional Tail Expectation (DCTE) against empirical estimates derived directly from historical returns. The analysis is conducted on a three-asset portfolio consisting of Bitcoin (BIT), Binance Coin (BNB), and Dogecoin (DOG), using an extreme Value-at-Risk (VaR) confidence level of $\alpha = 0.9975$.

Raw data of weekly price is first transformed into returns and plotted as shown in Figure 1. BIT and BNB shows a distinct positive linear relationship. The data points cluster along a diagonal, indicating that these two assets tend to move in the same direction. When one experiences positive returns, the other generally does as well, and they tend to decline together. But looking at DOG with both BIT and BNB, the positive correlation still exists, but with much heavier noise and less clustered than the BIT and BNB pair. DOG tends to follow the broader market trend but with significant deviations.

It is necessary to highlight the extreme volatility in DOG. The most prominent feature is the extreme downside volatility. In both the BIT vs. DOG and BNB vs. DOG plots, there are outlier points extending vertically downwards to approximately -3.0 on the DOG axis. These vertical drops correspond to weeks where DOG experienced massive losses while BIT and BNB saw only moderate declines. This suggests that DOG carries significant idiosyncratic risk and heavier tails, particularly on the downside compared to the other two assets.

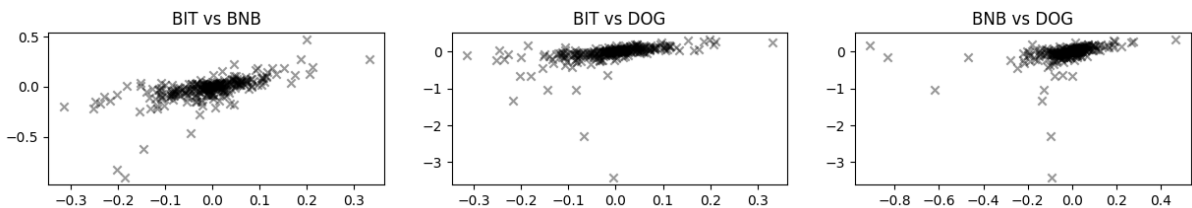


Figure 1: Weekly Returns

The marginal fit of the returns are plotted against the Student-t distribution as shown in Figure 2. BIT demonstrates an excellent fit as the data points follow the diagonal red line almost perfectly, staying well within the confidence interval. BNB also shows a good fit as the

data points generally follow the diagonal, but there is noticeable deviation at the extremes, particularly in the lower left quadrant. The fit is still strong with an $R^2 = 0.91$, although the model struggles with the asset's heavier tail compared to BIT. For the case of DOG, the lower left tail for extreme losses, the observed data points drop vertically away from the red line. This indicates that DOG experiences severe crashes far more frequently than the theoretical Student-t distribution accounts for. The $R^2 = 0.67$ confirms the poor statistical fit. This is aligned with the characteristics mentioned earlier with its high idiosyncratic risk and extreme heavy tail behavior. This could potentially lead to underestimation of downside risk for DOG.

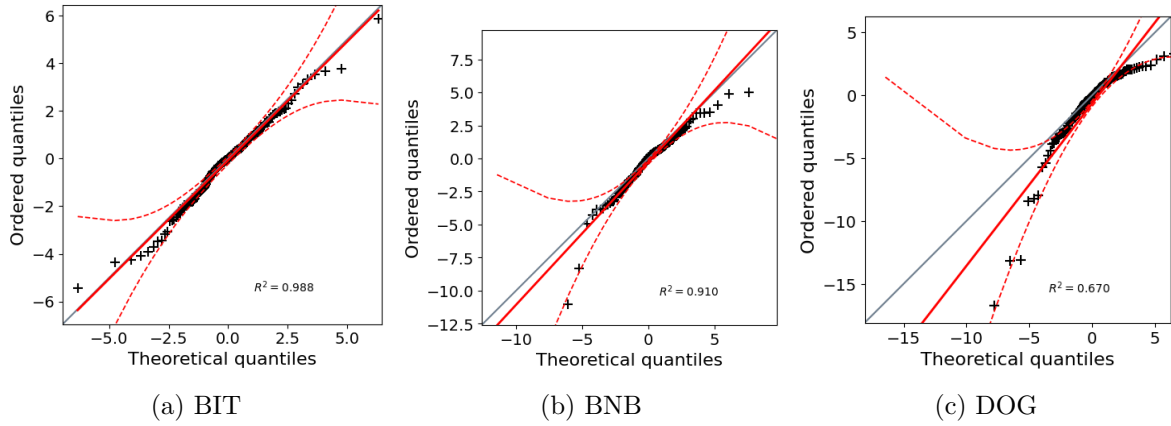


Figure 2: Marginal Fit

The return are then transformed into the exponential scale as shown in Figure 3 using the probability integral transform. For each $j = 1, \dots, d$, $X_j^E = \log(1 - F_j(X_j))$, where F_j is the j -th marginal of X . The values range from 0 to roughly 6. The fact that data points significantly populate the upper-right corners in all three plots validates the use of the SB procedure.

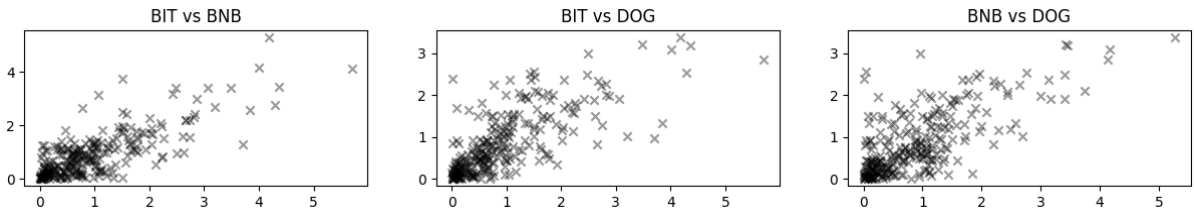


Figure 3: Exponential Returns

The excess is calculated by $Z = X^E - u^E \mid X^E \not\leq u^E$ and plotted as shown in Figure 4. The threshold $u^E = 0.83$ is chosen through visual inspections. The plot shows clustering around the threshold, suggesting it to be a suitable choice.

The excess are then simulated for 10000 samples in exponential scale. The QQ-plot compare the distribution of the simulated data $\tilde{X}$ against the underlying data X as shown in Figure 5. For BIT, the points follow the diagonal line tightly, indicating that the simulation replicates

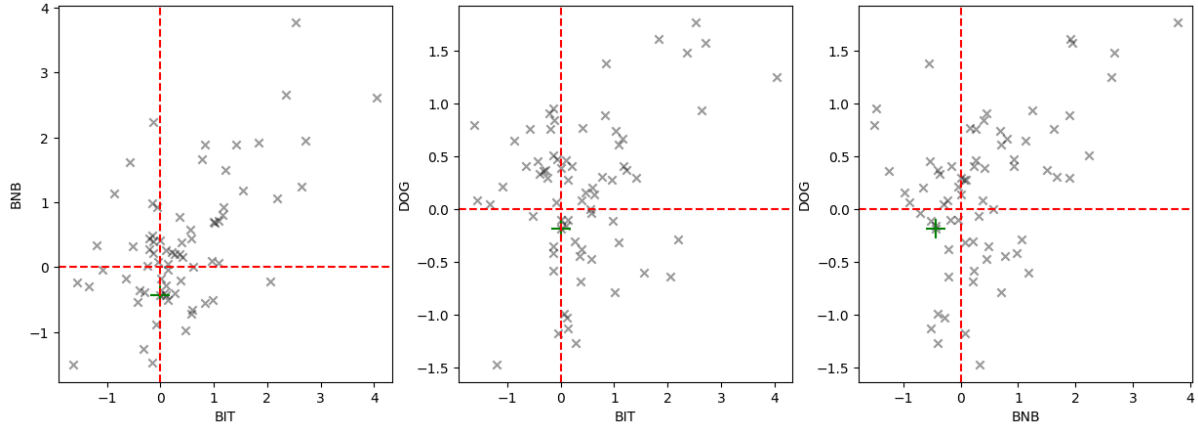


Figure 4: Excess Returns

BIT's risk distribution with high precision. For BNB, the points track the diagonal up to the extreme tail. There is a slight upward deviation at the very top, suggesting the simulation allows for slightly more severe extreme events than the historical sample alone. For DOG, the simulation captures the massive scale of DOG's tail risk. While BIT extends to 7, DOG extends to 20. The model is correctly simulating the fat tails specific to meme coins.

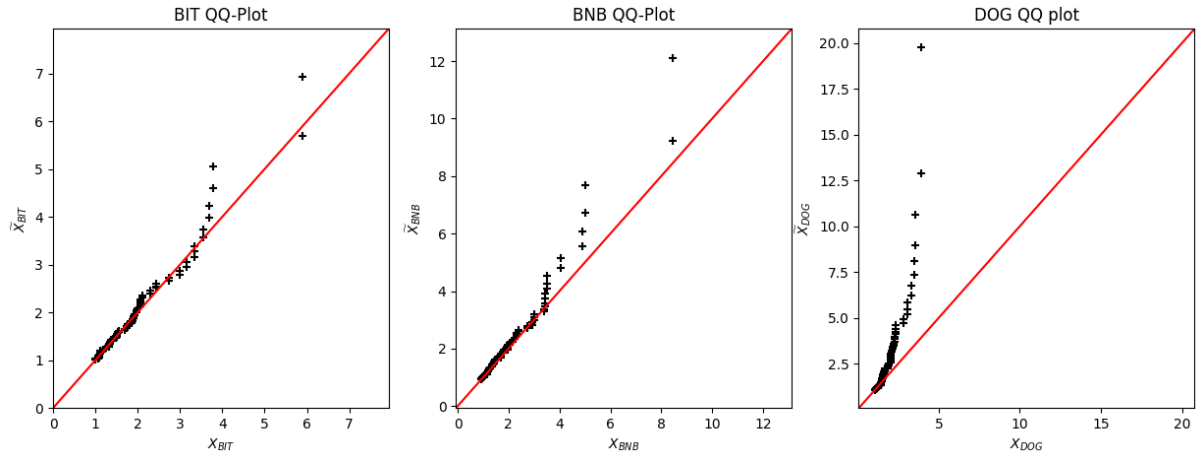


Figure 5: Simulated Returns

Figure 6 shows the comparison between the simulated data and the historical returns in exponential scale. In all three scatter plots, the red simulated points overlap with the black historical data points especially in the upper right quadrant, indicating the procedure's ability to replicate the tail behavior of the assets. But observing more closely to the BIT vs. BNB plot, the simulated data are heavily influenced by the volatility causing it to have a poorer coverage of the data clustering around moderate returns. The simulated data extend significantly further into the positive quadrants than the negative ones. This indicates the model is successfully generating Black Swan scenarios that are consistent with the portfolio's history but are more

severe than anything actually observed. In conclusion, the back-transformed simulation data provides a realistic and robust set of scenarios that capture both the day-to-day behavior of the market and the joint tail risks that are statistically possible but historically unobserved.

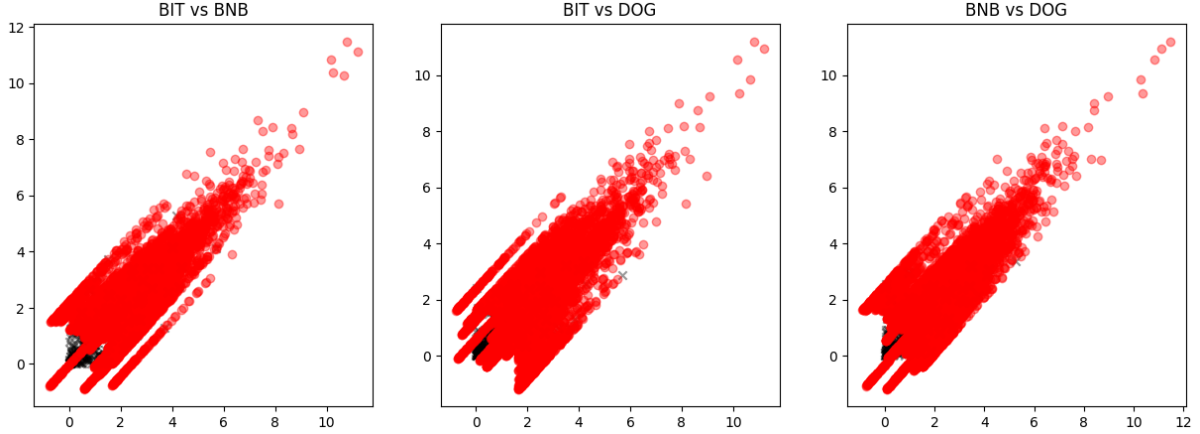


Figure 6: Simulated vs. Historical Returns

The simulated data are transformed back to its original scale using the inverse transformation $X_j^* = F_j^{-1} \left(1 - \exp(Z_j^* + u_j^E) \right)$, for $j = 1, \dots, d$. After transformation, the data are used for TRM estimation both from one-shot estimation and with iterations of 1000 runs for ES, MMES, and DCTE. ES is the expectation of observations larger than VaR and is defined as

$$ES_\alpha(X) = E[X|X > VaR_\alpha(X)] \quad (4)$$

The MMES is the risk associated with the target asset conditioned on a joint extreme event involving all other assets excluding the target asset, and is defined as

$$MMES_\alpha(X_j; X) = E[X_j | X_{-j} \geq v_{-j}^\alpha] \quad (5)$$

where $v_{-j}^\alpha = (VaR_\alpha(X_1), \dots, VaR_\alpha(X_d)) \in \mathbb{R}^d$. The DCTE is the risk associated with the target asset conditioned on all assets experiencing an extreme event, and is defined as

$$DCTE_\alpha(X_j; X) = E[X_j | X \geq v^\alpha] \quad (6)$$

Table 1 shows the one-shot estimation result. Empirical estimates returned NA due to the scarcity of historical observations exceeding the 99.75% threshold. The SB procedure demonstrates its ability to artificially populate the dataset with black swan data, allowing for the quantification of all risk measures even when there is insufficient historical data. The one-shot simulation establishes a clear hierarchy of risk among the assets, with DOG being the

most volatile, which is aligned with the historical performance of DOG and the characteristics of meme coins. Table 2 shows the statistics of the estimation results of 1000 iterations, demonstrating the robustness and the uncertainty of the estimates. The mean values from 1000 iterations align closely with the one-shot simulated results especially for BIT. But looking at the assets separately, DOG shows highly uncertain estimates with a standard deviation of 0.969 for DCTE. While we can be fairly confident in the risk estimates for BIT, the true tail risk for meme coins such as DOG fluctuates wildly and remains questionable.

Table 1: One-Shot Simulation Results

Stock	Metric	Empirical	Simulated
BIT	ES	(NA)	0.522
	MMES	(NA)	0.593
	DCTE	(NA)	0.640
BNB	ES	(NA)	1.378
	MMES	(NA)	1.517
	DCTE	(NA)	1.665
DOG	ES	(NA)	3.359
	MMES	(NA)	3.625
	DCTE	(NA)	4.195

Table 2: 1000 Iterations Simulation Results

Stock	Metric	Mean	Std. Dev.
BIT	ES	0.522	0.024
	MMES	0.586	0.042
	DCTE	0.625	0.045
BNB	ES	1.201	0.146
	MMES	1.325	0.214
	DCTE	1.472	0.247
DOG	ES	3.013	0.613
	MMES	3.119	0.739
	DCTE	3.773	0.969

5 Discussion

Overall, the SB procedure demonstrates its ability to artificially populate extreme data while preserving the multivariate tail dependency between assets. Its ability to facilitate the estimation of TRMs given lack of historical data especially in the multivariate setting provides means for risk management of asset portfolios. However, it is also important to point out the key assumptions underlying this bootstrapping method and how portfolios including high volatility stocks such as cryptocurrencies could influence the bias and consistency of the procedure.

The key assumption of tail dependency is a necessary condition for the procedure to work, thus, the correlation between the assets need to be tested before entering the procedure. If the portfolio contains highly diversified assets that do not exhibit joint behaviors during crashes, then it does not make sense to use this approach. Although the cryptocurrencies chosen for this simulation are of different profiles, i.e. they are different risk providers, within the same digital assets category, they are still under similar regimes and regulations, thus joint market crashes is not a surprise. The second assumption is that the Z vectors are iid, which might not be the case for high volatile assets such as cryptocurrencies. A GARCH model may need to be fitted first before entering the procedure to ensure the validity of the assumption. The third assumption is the student-t marginal assumption. Although this assumption is often true for traditional financial assets, highly volatile stocks often exhibit tail behaviors that are not symmetrical. This assumption should be questioned and potentially a skewed student-t distribution or a semi-parametric approach should be considered before the transformation to the exponential scale. If not, it could distort the input to the bootstrap, potentially biasing the TRMs.

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